HPC Project

Machine Learning Project

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1 Introduction and Data Presentation

In this project, we utilize the *Sleep Health and Lifestyle* dataset, which records health and lifestyle information for a sample of individuals. The variables in this dataset can be classified into three distinct categories:

1. Demographic and Professional Data: Unique Identifier, Age, Gender, and Occupation.
2. Lifestyle Indicators: Physical Activity Level, Daily Steps, Sleep Duration, and Sleep Quality (subjective scale).
3. Physiological Biomarkers: Body Mass Index (BMI), Heart Rate, Blood Pressure (in systolic/diastolic format), and Stress Level.

The dataset also contains a categorical target variable, Sleep Disorder (indicating the presence of insomnia, sleep apnea, or the absence of a disorder).

2 Problem Statement and Objectives

In the medical field, early detection of sleep disorders is crucial for preventing serious cardiovascular complications. The challenge is not only to predict correctly but, more importantly, not to miss pathological cases.

Our problem statement is as follows:
How can we maximize the detection of sleep disorders (Insomnia and Apnea) using lifestyle data and simple biomarkers, while minimizing the risk of false negatives?

Our objectives are:

- Identify the most influential factors on sleep health.
- Compare different supervised learning models (Logistic Regression, SVM, Random Forest, XGBoost).
- Optimize the selected model to maximize Recall, a critical metric in medical diagnosis.
- Evaluate the possibility of reducing the number of necessary variables without performance loss.

3 Methodological Approach

3.1 Data Preprocessing

We first cleaned and transformed the raw data:

- **Categorical Variable Encoding**:
  - *Occupation*: This variable is nominal (no logical order between "Engineer" and "Lawyer"). We used One-Hot Encoding to prevent the model from incorrectly interpreting a numerical hierarchy.
  - *BMI Category* and *Sleep Disorder*: These variables present an intrinsic progression of severity (e.g., Normal < Overweight < Obese). We opted for **Ordinal Encoding** to preserve this valuable ordinal information for prediction.
- **Blood Pressure Processing**: The initial variable was provided as a "Systolic/-Diastolic" string (e.g., "120/80"), which is mathematically unusable.
  - We split it into two distinct numerical variables: *Systolic BP* and *Diastolic BP*.
  - We then created a new synthetic variable, Pulse Pressure (Systolic - Diastolic). In cardiology, high pulse pressure is an indicator of arterial stiffness and a cardiovascular risk marker often correlated with sleep-disordered breathing.
- **Normalization (StandardScaler)**: The dataset variables have very heterogeneous magnitudes (e.g., *Daily Steps* $\approx$ 5000 vs *Sleep Duration* $\approx$ 7). Since some algorithms (like SVM or KNN) are based on Euclidean distance calculations, large-scale variables would artificially dominate the decision. We therefore standardized all numerical variables (mean = 0, standard deviation = 1) to ensure an equitable contribution from each factor.

3.2 Exploratory Analysis

Exploratory Data Analysis (EDA) is a fundamental step to understand the dataset structure and relationships between variables before modeling.
First, we examined the distribution of the target variable (Sleep Disorders). As illustrated in Figure 1, the dataset shows a class imbalance: approximately 60% of individuals have no disorder, while sleep apnea and insomnia each represent about 20% of observations. This imbalance must be taken into account during model evaluation, justifying the use of adapted metrics such as Recall rather than simple Accuracy.

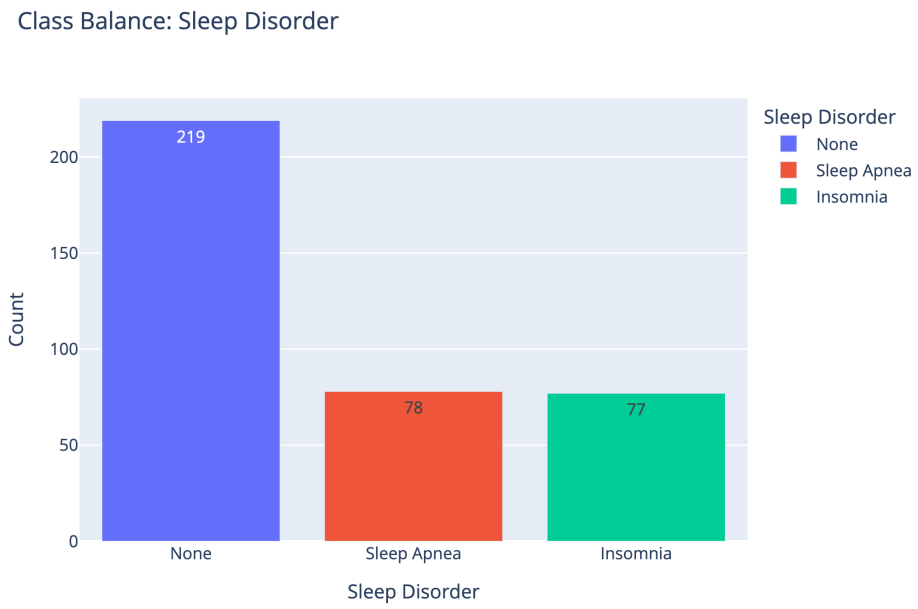

Figure 1: Distribution of sleep disorder classes

Next, we studied the linear dependencies between variables using a correlation matrix (Figure 2). This analysis highlights several strong relationships. Notably, we observe a marked negative correlation between stress level and sleep quality, as well as with sleep duration. Furthermore, sleep duration and quality are themselves strongly positively correlated. Regarding pathologies, sleep apnea shows a significant correlation with Body Mass Index (BMI) and blood pressure measures (systolic and diastolic), which is consistent with medical literature.

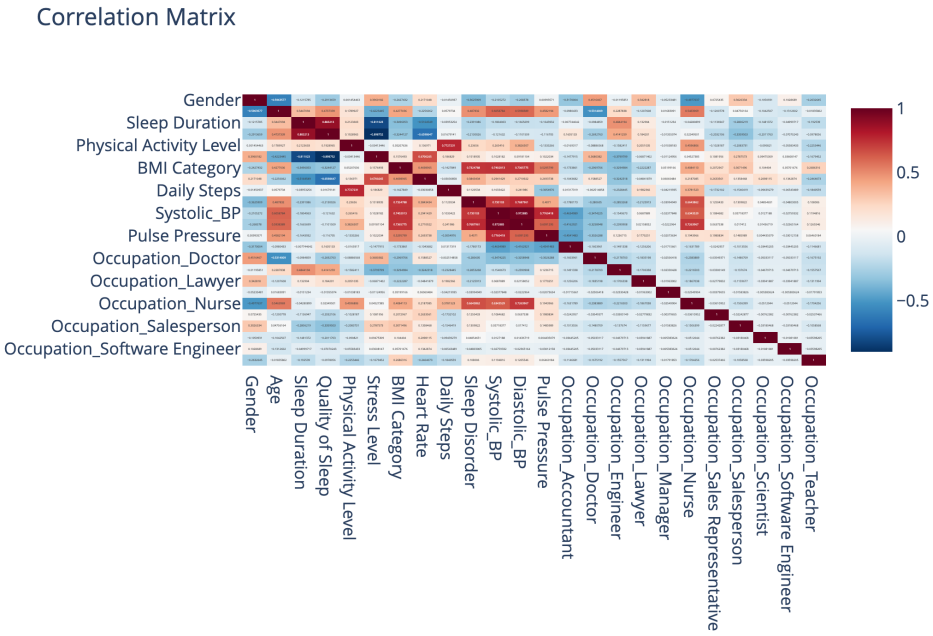

Figure 2: Correlation matrix of variables

To deepen our understanding of discriminating factors, we used a Random Forest algorithm to evaluate the relative importance of variables (Feature Importance). As shown in Figure 3, systolic and diastolic blood pressures appear as the most influential predictors, confirming the observations from the correlation matrix.

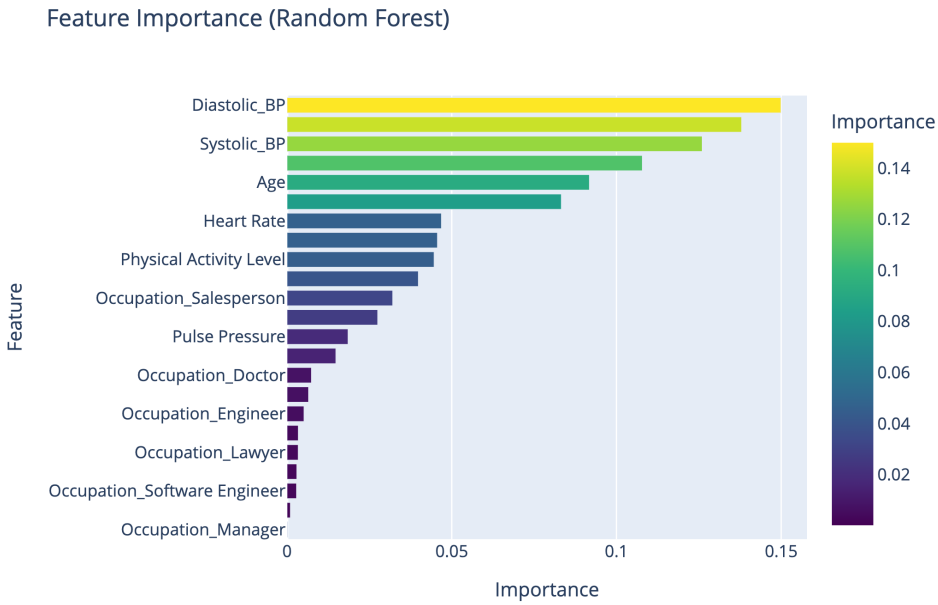

Figure 3: Feature importance via Random Forest

Finally, the analysis of age distribution according to sleep disorders (Figure 4) reveals

interesting demographic trends. Healthy individuals are, on average, the youngest. We observe a shift in average age for people suffering from insomnia, and even more so for those with sleep apnea, who constitute the oldest group on average. This observation suggests that age is an increasing risk factor for these pathologies.

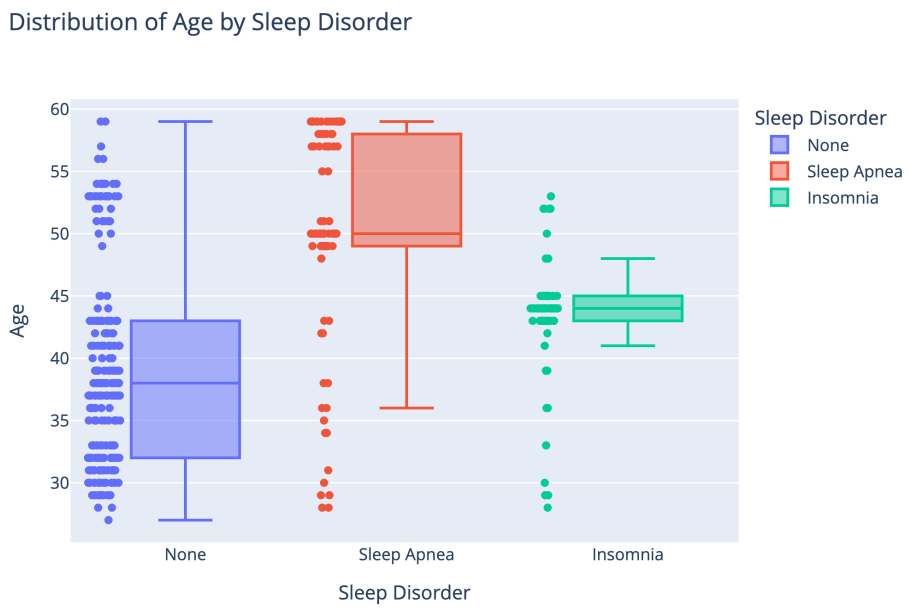

Figure 4: Age distribution by sleep disorder category

3.3 Evaluation Protocol and Metrics

To ensure the robustness of our results against class imbalance, we adopted a Stratified Cross-Validation strategy with $k = 5$. This method ensures that the proportion of each class (Normal, Insomnia, Apnea) is preserved in each training and test fold.
The preferred performance metric is **Macro Recall**. In a medical diagnostic context, the absolute priority is to minimize False Negatives, i.e., to avoid classifying a patient suffering from a pathology as "healthy". We also examine Confusion Matrices to finely analyze classification errors between different pathologies.

3.4 Candidate Models

We selected a diverse panel of algorithms to cover different learning approaches:

- **Logistic Regression**: For its simplicity and interpretability (linear model).
- **Support Vector Machine (SVM)**: For its ability to find complex separating hyperplanes in high-dimensional spaces.
- **Random Forest**: An ensemble method robust to overfitting and capable of capturing non-linearities.

- **XGBoost**: A Gradient Boosting method recognized for its high performance on structured tabular data.

3.5 Experimental Strategy

Our selection process took place in three iterative phases, guided by a desire for simplification and clinical applicability:

1. **Phase 1: Baseline Evaluation**
   Initially, we trained all models on all available variables. The objective was to establish a theoretical maximum performance level without variable selection constraints.

2. **Phase 2: Noise Elimination (Exclusion of Occupation)**
   Our preliminary analyses suggesting that profession (*Occupation*) did not have a strong causal link with sleep disorders, we tested the models without these variables. The hypothesis was that these features could introduce noise or Overfitting without bringing real predictive gain.

3. **Phase 3: "Clinical Biomarkers" Approach (Minimalist)**
   From a perspective of real-world deployment and diagnostic cost reduction, we sought to determine if a high-performing model could be built solely from 5 key biomarkers: *Age, Sleep Duration, Systolic BP, Diastolic BP, and BMI Category.* This approach aims to propose a lightweight pre-diagnostic tool, not requiring a complex lifestyle questionnaire.

4 Results and Discussion

This section details the performance obtained during the three phases of our experimental strategy. The objective is to validate the relevance of dimensionality reduction.

4.1 Phase 1: Baseline (All variables)

Table 1 presents the results obtained using all dataset variables.

Model	Accuracy	Recall (Macro)	Precision (Macro)	F1 Score (Macro)
Logistic Regression	0.8986	0.8722	0.8788	0.8726
SVM	0.9014	0.8652	0.8796	0.8690
Random Forest	**0.9127**	**0.8862**	**0.8992**	**0.8896**
XGBoost	0.9099	0.8811	0.8933	0.8849

Table 1: Performance with all variables

In this reference configuration, Random Forest achieves the best overall performance with a Recall of 0.8862.

4.2 Phase 2: Removal of Occupation

By removing profession-related variables, we observe the following results (Table 2):

Model	Accuracy	Recall (Macro)	Precision (Macro)	F1 Score (Macro)
Logistic Regression	0.9014	0.8738	0.8887	0.8778
SVM	0.9014	0.8649	0.8821	0.8710
Random Forest	**0.9155**	**0.8878**	**0.9034**	**0.8923**
XGBoost	0.9127	0.8827	0.8974	0.8875

Table 2: Performance without Occupation variables

Performances remain stable, or even slightly improve for Random Forest and XGBoost. This confirms our hypothesis that profession is not an essential predictor and can be removed without information loss.

4.3 Phase 3: Minimalist Biomarkers

Finally, we evaluate the models on the restricted subset of 5 biomarkers (Table 3).

Model	Accuracy	Recall (Macro)	Precision (Macro)	F1 Score (Macro)
Logistic Regression	0.8563	0.8071	0.8315	0.8088
SVM	0.8930	0.8573	0.8828	0.8661
Random Forest	0.9070	0.8735	0.8949	0.8815
XGBoost	**0.9127**	**0.8827**	**0.8995**	**0.8886**

Table 3: Performance with minimalist biomarkers

4.4 Comparative Analysis and Model Choice

Comparing the three phases reveals a remarkable result: moving from a full dataset to only 5 biomarkers results in negligible performance loss, or even none for some models.
The **XGBoost** model, in particular, maintains a Recall of **0.8827** with the biomarkers, which is a score almost identical to its performance with all variables (0.8811) and very close to the best model from Phase 1 (Random Forest: 0.8862).
This robustness of XGBoost to variable reduction, combined with its execution speed, motivates our choice to retain it as the final model. The "Minimalist Biomarkers" approach is therefore validated: it offers equivalent diagnostic accuracy while being much simpler to deploy in practice.

4.5 Hyperparameter Optimization

To further maximize the model's sensitivity (Recall), we proceeded with an exhaustive search for the best hyperparameters (Grid Search) for XGBoost. We explored different combinations of *learning_rate*, *max_depth*, *n_estimators*, and *subsample*.

The best identified configuration is as follows:

- *learning_rate* : 0.05
- *max_depth* : 4
- *n_estimators* : 100
- *subsample* : 1.0

This optimization improved the cross-validation score, bringing the average Macro Recall to **0.8967**. This confirms the value of fine-tuning the model to capture the subtleties of decision boundaries between classes.

4.6 Exploration of Ensemble Methods

In an effort to push model performance to the maximum, we explored Ensemble Learning techniques. The idea was to combine the predictions of our four base models (Logistic Regression, SVM, Random Forest, XGBoost) to hope to reduce variance and improve generalization.
We implemented three voting strategies:

- **Majority Vote (Hard Voting)**: The majority class among predictions wins.
- **Probability Averaging (Soft Voting)**: Average of probabilities predicted by each model.
- **Weighted Voting**: Weighted vote granting more importance to the best-performing models (Random Forest and XGBoost).

However, the results obtained were not conclusive. The best score obtained (via Soft Voting) caps at a Recall of 0.8783, which remains lower than the performance of our optimized XGBoost model alone (0.8967). The added complexity inherent in maintaining four models in parallel not being justified by a performance gain, we confirmed our choice to keep the single XGBoost model for the final solution.

4.7 Final Validation

To evaluate the real generalization capacity of our solution, we applied the optimized model to the validation dataset (a sample never seen during training or optimization).
The final results are excellent:

- **Accuracy** : 0.9474
- **Recall (Macro)** : 0.9697
- **Precision (Macro)** : 0.9333

- **F1 Score (Macro)** : 0.9471

The confusion matrix below details the predictions:

	Predicted: None	**Predicted: Insomnia**	**Predicted: Apnea**
Actual: None	10	0	1
Actual: Insomnia	0	4	0
Actual: Apnea	0	0	4

Table 4: Confusion Matrix on the validation set

The analysis of the confusion matrix reveals a crucial point for our medical problem: **no pathological case (Insomnia or Apnea) was missed**. The 4 cases of insomnia and 4 cases of apnea were correctly detected (0 False Negatives). The only error committed is a False Positive (a healthy individual classified as suffering from apnea), which is acceptable in a screening context where a false alarm is preferred to a missed diagnosis.

5 Conclusion

This project demonstrated that it is possible to predict sleep disorders (Insomnia and Apnea) with high reliability using a very limited number of physiological biomarkers (Age, BMI, Blood Pressure, Sleep Duration).
Our approach, based on an optimized XGBoost model, achieves a Recall of nearly 97% on validation data, perfectly fulfilling the objective of minimizing false negatives. The simplicity of the required data (5 variables) makes this model easily integratable into connected health tools or rapid pre-diagnostic protocols, thus offering a concrete solution for improving sleep disorder screening.